

Practice Set 5

MA628 - Financial Derivatives Pricing

February 16, 2026

1. A stock price is currently at 50 Rs. Over each of the next two 3-months periods it is expected to go up by 6% or down by 5%. The risk free rate is 5% per annum compounded quarterly. Find the value of a 6-months European call option with strike 51 Rs.
2. A stock price is currently at 50 Rs. Over each of the next two 3-months periods it is expected to go up by 6% or down by 5%. The risk free rate is 5% per annum compounded continuously. Find the value of a 6-month American put option with strike 50 Rs.
3. A stock price is currently at 50 Rs. Over each of the next two 3-months periods it is expected to go up by 6% or down by 5%. The risk free rate is 5% per annum compounded quarterly. Find the value of a 6-month American call option with strike 50 Rs. Show by direct calculation that American call and European call option prices are same.
4. Consider a one step Binomial market model for a stock which pays a dividend with a continuous dividend yield q . Money market account is with constant rate of interest r (per annum) compounded continuously. If the above market model is arbitrage free, then show that $0 < d < e^{r-q} < u$.
5. A stock price is currently at 50 Rs. Over each of the next two 3-months periods it is expected to go up by 6% or down by 5%. The risk free rate is 5% per annum compounded quarterly. For a 6-month European call option with strike 50 Rs, find the values of Greeks Δ , Γ and Θ .
6. A stock is currently selling for \$50. The stock price could go up by 10% or fall by 5% each month. The monthly interest rate is 1% (periodic rate). Calculate the price of a European call option on the stock with an exercise price of \$50 and a maturity of two months. (use the two period binomial method)
7. A stock is currently selling for \$50. The stock price could go up by 10% or fall by 5% each month. The monthly interest rate is 1% (periodic rate). Calculate the price of a European put option on the stock with an exercise price of \$55 and a maturity of two months. (use the two-stage binomial method)